EASTERN INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTING
AND
INFORMATION TECHNOLOGY

Week 7 Practice Assignment – Quarter 2, 2025-2026
Method: LAB 6 Practice 6
Course Name: Advanced Databases
Course Code: CSW 436
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Important Instructions –

- Implement each question using MongoDB.
- find(), update(), delete().
- Use $group, $count, $avg, $max and other NoSQL method for query processing and optimization.
- Document all question Output/results and NoSQL code for every question in a Word / PDF document and save it.
- After completing all questions, upload the document to Moodle.

Topic: MongoDB (CRUD, schema design)

1. **Create a database named schoolDB and a collection called students. Insert 5 student records with the following fields:**
   student_id, name, class, age, marks, city

   Asnwer:

```
db.students.insertMany([
  { student_id: 101, name: "Zhuang Fangyi", class: "10A", age: 16, marks: 90, city: "Ho Chi Minh" },
  { student_id: 102, name: "Perlica", class: "10B", age: 15, marks: 72, city: "Binh Duong" },
  { student_id: 103, name: "Chen Qianyu", class: "10A", age: 16, marks: 65, city: "Dong Nai" },
  { student_id: 104, name: "Antal", class: "10B", age: 15, marks: 85, city: "Binh Duong" },
  { student_id: 105, name: "Nefarith", class: "10A", age: 16, marks: 30, city: "Ho Chi Minh" }
])
```

```
_id: ObjectId('69a4e972745b1a5d5ce0150f')
student_id : 101
name : "Zhuang Fangyi"
class : "10A"
age : 16
marks : 90
city : "Ho Chi Minh"
```

```
_id: ObjectId('69a4e972745b1a5d5ce01510')
student_id : 102
name : "Perlica"
class : "10B"
age : 15
marks : 72
city : "Binh Duong"
```

```
_id: ObjectId('69a4e972745b1a5d5ce01511')
student_id : 103
name : "Chen Qianyu"
class : "10A"
age : 16
marks : 65
city : "Dong Nai"
```

```
_id: ObjectId('69a4e972745b1a5d5ce01512')
student_id : 104
name : "Antal"
class : "10B"
age : 15
marks : 85
city : "Binh Duong"
```

```
_id: ObjectId('69a4e972745b1a5d5ce01513')
student_id : 105
name : "Nefarith"
class : "10A"
age : 16
marks : 30
city : "Ho Chi Minh"
```

2. **Write queries to:**
   - **Display all students.**
   - **Display students who scored more than 70 marks.**
   - **Display students from class "10A".**

Asnwer:

- **Display all students.**

db.students.find()

```
> db.students.find()
< {
    _id: ObjectId('69a4e972745b1a5d5ce0150f'),
    student_id: 101,
    name: 'Zhuang Fangyi',
    class: '10A',
    age: 16,
    marks: 90,
    city: 'Ho Chi Minh'
  }
  {
    _id: ObjectId('69a4e972745b1a5d5ce01510'),
    student_id: 102,
    name: 'Perlica',
    class: '10B',
    age: 15,
    marks: 72,
    city: 'Binh Duong'
  }
  {
    _id: ObjectId('69a4e972745b1a5d5ce01511'),
    student_id: 103,
    name: 'Chen Qianyu',
    class: '10A',
    age: 16,
    marks: 65,
```

```
{
  _id: ObjectId('69a4e972745b1a5d5ce01511'),
  student_id: 103,
  name: 'Chen Qianyu',
  class: '10A',
  age: 16,
  marks: 65,
  city: 'Dong Nai'
}
{
  _id: ObjectId('69a4e972745b1a5d5ce01512'),
  student_id: 104,
  name: 'Antal',
  class: '10B',
  age: 15,
  marks: 85,
  city: 'Binh Duong'
}
{
  _id: ObjectId('69a4e972745b1a5d5ce01513'),
  student_id: 105,
  name: 'Nefarith',
  class: '10A',
  age: 16,
  marks: 30,
  city: 'Ho Chi Minh'
}
```

Display students who scored more than 70 marks.
db.students.find({marks:{$gt:70}})

```
> db.students.find({marks:{$gt:70}})
< {
    _id: ObjectId('69a4e972745b1a5d5ce0150f'),
    student_id: 101,
    name: 'Zhuang Fangyi',
    class: '10A',
    age: 16,
    marks: 90,
    city: 'Ho Chi Minh'
  }
  {
    _id: ObjectId('69a4e972745b1a5d5ce01510'),
    student_id: 102,
    name: 'Perlica',
    class: '10B',
    age: 15,
    marks: 72,
    city: 'Binh Duong'
  }
```

```
    city: 'Binh Duong'
  }
  {
    _id: ObjectId('69a4e972745b1a5d5ce01512'),
    student_id: 104,
    name: 'Antal',
    class: '10B',
    age: 15,
    marks: 85,
    city: 'Binh Duong'
  }
```

Display students from class "10A".

db.students.find({ class: "10A" })

```
> db.students.find({ class: "10A" })
< {
    _id: ObjectId('69a4e972745b1a5d5ce0150f'),
    student_id: 101,
    name: 'Zhuang Fangyi',
    class: '10A',
    age: 16,
    marks: 90,
    city: 'Ho Chi Minh'
  }
  {
    _id: ObjectId('69a4e972745b1a5d5ce01511'),
    student_id: 103,
    name: 'Chen Qianyu',
    class: '10A',
    age: 16,
    marks: 65,
    city: 'Dong Nai'
  }
  {
    _id: ObjectId('69a4e972745b1a5d5ce01513'),
    student_id: 105,
    name: 'Nefarith',
    class: '10A',
    age: 16,
    marks: 30,
    city: 'Ho Chi Minh'
  }
```

3. **Write queries to:**
   - **Update the marks of the student whose student_id is 101.**
   - **Increase marks by 5 for all students in class "10A".**

Asnwer:

- **Update the marks of the student whose student_id is 101.**

```
db.students.updateOne(
  { student_id: 101 },
  { $set: { marks: 95 } }
)
```

```
> db.students.updateOne(
    { student_id: 101 },
    { $set: { marks: 95 } }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

```
_id: ObjectId('69a4e972745b1a5d5ce0150f')
student_id : 101
name : "Zhuang Fangyi"
class : "10A"
age : 16
marks : 95
city : "Ho Chi Minh"
```

- **Increase marks by 5 for all students in class "10A".**

```
db.students.updateMany(
  { class: "10A" },
  { $inc: { marks: 5 } }
)
```

```
> db.students.updateMany(
    { class: "10A" },
    { $inc: { marks: 5 } }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 3,
    modifiedCount: 3,
    upsertedCount: 0
  }
```

```
_id: ObjectId('69a4e972745b1a5d5ce0150f')
student_id : 101
name : "Zhuang Fangyi"
class : "10A"
age : 16
marks : 100
city : "Ho Chi Minh"

_id: ObjectId('69a4e972745b1a5d5ce01511')
student_id : 103
name : "Chen Qianyu"
class : "10A"
age : 16
marks : 70
city : "Dong Nai"

_id: ObjectId('69a4e972745b1a5d5ce01513')
student_id : 105
name : "Nefarith"
class : "10A"
age : 16
marks : 35
city : "Ho Chi Minh"
```

4. **Write queries to:**
   - **Delete the student whose student_id is 105.**
   - **Delete students who scored less than 35 marks.**

Asnwer:

- **Delete the student whose student_id is 105.**

db.students.deleteOne({ student_id: 105 })

```
> db.students.deleteOne({ student_id: 105 })
< {
    acknowledged: true,
    deletedCount: 1
  }
```

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- **Delete students who scored less than 35 marks.**

Insert for delete:

db.students.insertMany([
  { student_id: 106, name: "random guy 1", class: "10A", age: 16, marks: 10, city: "Ho Chi Minh" },
  { student_id: 107, name: "random guy 2", class: "10B", age: 15, marks: 22, city: "Binh Duong" }
])

```
> db.students.insertMany([
    { student_id: 106, name: "random guy 1", class: "10A", age: 16, marks: 10, city: "Ho Chi Minh" },
    { student_id: 107, name: "random guy 2", class: "10B", age: 15, marks: 22, city: "Binh Duong" }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('69a4f299745b1a5d5ce01516'),
      '1': ObjectId('69a4f299745b1a5d5ce01517')
    }
  }
```

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db.students.deleteMany({ marks: { $lt: 35 } })

```
> db.students.deleteMany({ marks: { $lt: 35 } })
< {
    acknowledged: true,
    deletedCount: 2
  }
```

1 – 4 of 4

5. Perform All CRUD Operations for the following:
   - Insert a new student record.
   - Update the student's marks.
   - Display the updated record.
   - Delete the same student record.

Asnwer:

- **Insert a new student record.**

db.students.insertOne({

student_id: 108,

name: "random guy",

class: "10B",

age: 15,

marks: 75,

city: "Ho Chi Minh"

})

```
> db.students.insertOne({
    student_id: 108,
    name: "random guy",
    class: "10B",
    age: 15,
    marks: 75,
    city: "Ho Chi Minh"
  })
< {
    acknowledged: true,
    insertedId: ObjectId('69a4f40d745b1a5d5ce01518')
  }
```

```
_id: ObjectId('69a4f40d745b1a5d5ce01518')
student_id : 108
name : "random guy"
class : "10B"
age : 15
marks : 75
city : "Ho Chi Minh"
```

- **Update the student's marks.**

db.students.updateOne(

{ student_id: 108},

{ $set: { marks: 90 } }

)

```
> db.students.updateOne(
    { student_id: 108},
    { $set: { marks: 90 } }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

```
_id: ObjectId('69a4f40d745b1a5d5ce01518')
student_id : 108
name : "random guy"
class : "10B"
age : 15
marks : 90
city : "Ho Chi Minh"
```

- **Display the updated record.**

db.students.find({ student_id: 108})

```
> db.students.find({ student_id: 108})
< {
    _id: ObjectId('69a4f40d745b1a5d5ce01518'),
    student_id: 108,
    name: 'random guy',
    class: '10B',
    age: 15,
    marks: 90,
    city: 'Ho Chi Minh'
  }
```

- **Delete the same student record.**

db.students.deleteOne({ student_id: 108})

```
> db.students.deleteOne({ student_id: 108})
< {
    acknowledged: true,
    deletedCount: 1
  }
```

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OLAP Operations (Aggregation in MongoDB): Use the students collection for the following aggregation queries.

1. Use - Group By: write an aggregation query to calculate the average marks of students in each class.
2. Sum of Marks: write an aggregation query to calculate the total marks of students in each class.

3. Write an aggregation query to count the number of students in each city.
4. Filter and Group - write an aggregation query to:
   - Find the average marks of students from the city "Binh Doung".
5. Sort and top results - write an aggregation query to:
   - Display classes sorted by average marks in descending order.

Asnwer:

Use - Group By: write an aggregation query to calculate the average marks of students in each class.

```
db.students.aggregate([
  {
    $group: {
      _id: "$class", averageMarks: { $avg: "$marks" }
    }
  }
])
```

```
> db.students.aggregate([
    {
      $group: {
        _id: "$class", averageMarks: { $avg: "$marks" }
      }
    }
  ])
< {
    _id: '10A',
    averageMarks: 85
  }
  {
    _id: '10B',
    averageMarks: 78.5
  }
```

Sum of Marks: write an aggregation query to calculate the total marks of students in each class.

```
db.students.aggregate([
  {
    $group: {
      _id: "$class", totalMarks: { $sum: "$marks" }
    }
  }
])
```

```
> db.students.aggregate([
    {
      $group: {
        _id: "$class", totalMarks: { $sum: "$marks" }
      }
    }
  ])
< {
    _id: '10B',
    totalMarks: 157
  }
  {
    _id: '10A',
    totalMarks: 170
  }
```

Write an aggregation query to count the number of students in each city.

```
db.students.aggregate([
  {
    $group: {
      _id: "$city", studentCount: { $count: {} }
    }
  }
])
```

```
> db.students.aggregate([
    {
      $group: {
        _id: "$city", studentCount: { $count: {} }
      }
    }
  ])
< {
    _id: 'Dong Nai',
    studentCount: 1
  }
  {
    _id: 'Ho Chi Minh',
    studentCount: 1
  }
  {
    _id: 'Binh Duong',
    studentCount: 2
  }
```

Filter and Group - write an aggregation query to:

- **Find the average marks of students from the city "Binh Doung".**

```
db.students.aggregate([
  {
    $match: { city: "Binh Duong" }
  },
  {
    $group: {
      _id: "$city", averageMarks: { $avg: "$marks" }
    }
  }
])
```

```
  }
> db.students.aggregate([
    {
      $match: { city: "Binh Duong" }
    },
    {
      $group: {
        _id: "$city", averageMarks: { $avg: "$marks" }
      }
    }
  ])
< {
    _id: 'Binh Duong',
    averageMarks: 78.5
  }
```

Sort and top results - write an aggregation query to:

- **Display classes sorted by average marks in descending order.**

```
db.students.aggregate([
  {
    $group: {
      _id: "$class", averageMarks: { $avg: "$marks" }
    }
  },
  {
    $sort: { averageMarks: -1 }
  }
])
```

```
> db.students.aggregate([
    {
      $group: {
        _id: "$class", averageMarks: { $avg: "$marks" }
      }
    },
    {
      $sort: { averageMarks: -1 }
    }
  ])
< {
    _id: '10A',
    averageMarks: 85
  }
  {
    _id: '10B',
    averageMarks: 78.5
```

Migrating from mysql to mongoDb

Mysql:

```sql
create database shopdb;
use shopdb;

CREATE TABLE customers (
id INT PRIMARY KEY,
name VARCHAR(100),
email VARCHAR(100)
);
CREATE TABLE orders (
id INT PRIMARY KEY,
customer_id INT,
order_date DATE,
total DECIMAL(10,2),
FOREIGN KEY (customer_id) REFERENCES customers(id)
);

-- Insert sample customers
INSERT INTO customers (id, name, email) VALUES
(1, 'John Doe', 'john.doe@email.com'),
(2, 'Jane Smith', 'jane.smith@email.com'),
(3, 'Bob Johnson', 'bob.johnson@email.com'),
```

Command line

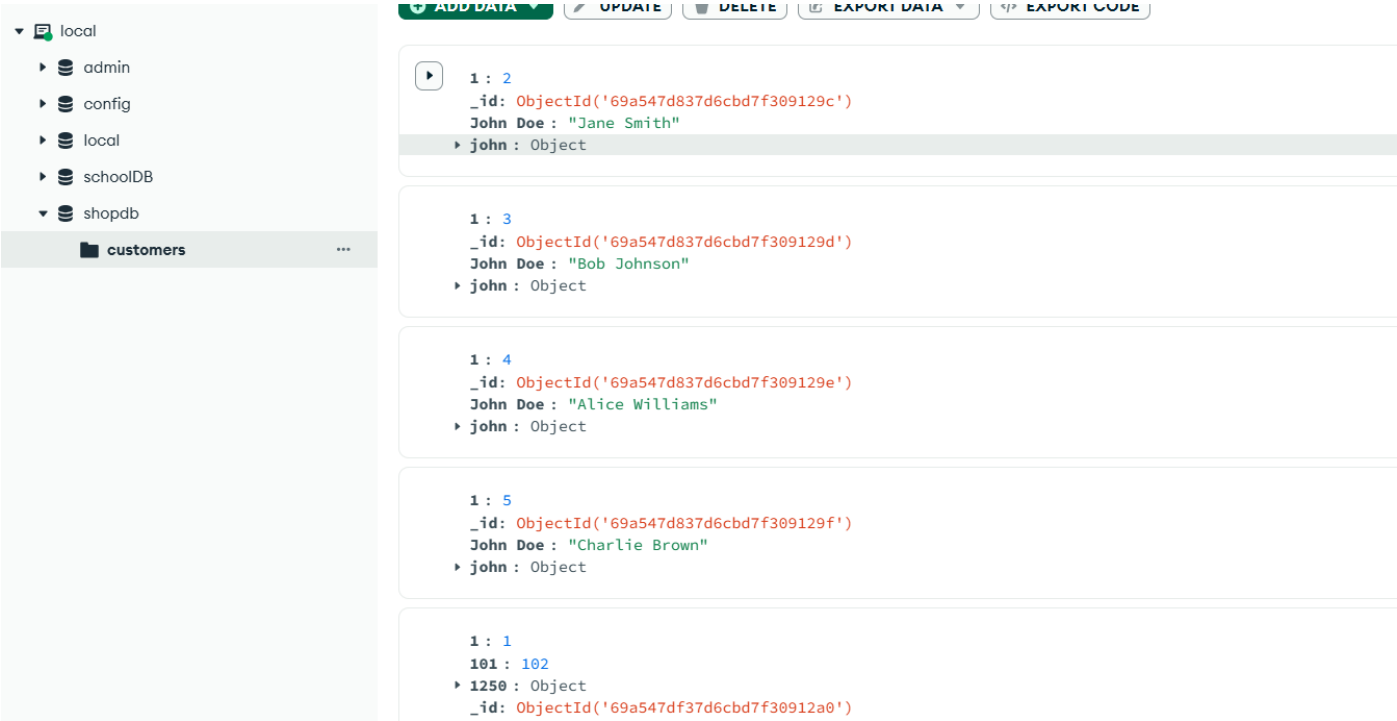
```
C:\Users\Giahu>mysql -u root -p
```

```
SHOW VARIABLES LIKE 'secure_file_priv';

SELECT * FROM customers
INTO OUTFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/customers.csv'
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n';

SELECT * FROM orders
INTO OUTFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/orders.csv'
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n';

exit
```

```
copy "C:\ProgramData\MySQL\MySQL Server 8.0\Uploads\customers.csv" C:\Users\Giahu\

copy "C:\ProgramData\MySQL\MySQL Server 8.0\Uploads\orders.csv" C:\Users\Giahu\

cd C:\Users\Giahu

dir *.csv

type customers.csv

type orders.csv
```

```
mongoimport --db shopdb --collection customers --type csv --headerline --file customers.csv

mongoimport --db shopdb --collection orders --type csv --headerline --file orders.csv
```

MongoDB


ADD DATA
UPDATE
DELETE
EXPORT DATA
EXPORT CODE
local
admin
config
local
schoolDB
shopdb
customers
1 : 2
_id: ObjectId('69a547d837d6cbd7f309129c')
John Doe : "Jane Smith"
john : Object
1 : 3
_id: ObjectId('69a547d837d6cbd7f309129d')
John Doe : "Bob Johnson"
john : Object
1 : 4
_id: ObjectId('69a547d837d6cbd7f309129e')
John Doe : "Alice Williams"
john : Object
1 : 5
_id: ObjectId('69a547d837d6cbd7f309129f')
John Doe : "Charlie Brown"
john : Object
1 : 1
101 : 102
1250 : Object
_id: ObjectId('69a547df37d6cbd7f30912a0')

```
1 : 1
101 : 102
▸ 1250 : Object
_id: ObjectId('69a547df37d6cbd7f30912a0')
2024-01-15 : 2024-02-20T00:00:00.000+00:00
```

```
1 : 2
101 : 103
▸ 1250 : Object
_id: ObjectId('69a547df37d6cbd7f30912a1')
2024-01-15 : 2024-01-18T00:00:00.000+00:00
```

```
1 : 3
101 : 104
▸ 1250 : Object
_id: ObjectId('69a547df37d6cbd7f30912a2')
2024-01-15 : 2024-01-22T00:00:00.000+00:00
```

```
1 : 2
101 : 105
▸ 1250 : Object
_id: ObjectId('69a547df37d6cbd7f30912a3')
2024-01-15 : 2024-02-10T00:00:00.000+00:00
```

```
2024-01-15 : 2024-03-01T00:00:00.000+00:00
```

```
1 : 5
101 : 108
1250 : Object
_id: ObjectId('69a547df37d6cbd7f30912a6')
2024-01-15 : 2024-02-28T00:00:00.000+00:00
```

```
1 : 3
101 : 109
1250 : Object
_id: ObjectId('69a547df37d6cbd7f30912a7')
2024-01-15 : 2024-03-05T00:00:00.000+00:00
```

```
1 : 4
101 : 110
1250 : Object
_id: ObjectId('69a547df37d6cbd7f30912a8')
2024-01-15 : 2024-03-10T00:00:00.000+00:00
```

Lecturer
Rohit Kumar Kasera